

# Meg Montañez Davenport

DECENTRALIZED SYSTEMS ARCHITECT  
WEB3 ENGINEER · APPLIED RESEARCHER

## CONTACT

- future.systems.lab@proton.me
- sovereigncoverage@futuresystemslab.io
- NC, USA — Remote
- linkedin.com/in/dr-meg-montanez-davenport
- github.com/Future-Systems-Lab/Future-Systems-Lab-profile
- americaoutloud.news/author/meg-montanez-davenport

## BLOCKCHAIN & WEB3

SolidityHardhatRemix IDE

ERC-20/721Brave WalletEIP-6963

window.ethereumDIDs / SSIIPF5

FilecoinMerkle DAGsZK Proofs

libp2pBlockscout

## ENGINEERING & DATA

Next.jsPostgreSQL 17PythonFlask

Streamlit systemdREST APIsVercel

SSH Ed25519Git CI/CDAgile/Scrum

Tableau

## HEALTHCARE & INFORMATICS

HL7FHIROMOPHIPAAEHR Mapping

Clinical Data ModelsOrthomolecular

## ORGANIZATIONS

- INCOSE — Systems Engineering
- AMIA — Medical Informatics
- HIMSS — Health IT
- ISC² — Cybersecurity
- PDMA — Product Management
- ISOM — Orthomolecular Medicine
- NANP — Nutrition Professionals

## CERTIFICATIONS

- Certified Blockchain Healthcare Professional (CBHP) — Blockchain Council · Apr 2025
- Certified Smart Contract AuditorBlockchain Council · Jan 2026
- Health Informatics — Johns HopkinsMay 2025
- Web3 & Blockchain — INSEADApr 2025
- CISA — PacktAug 2025
- SSI Master — DizmeliDMar 2025

## SUMMARY

I design and deploy decentralized infrastructure for sovereign data governance — using regulated behavioral health as a deliberate proving ground for architectures that scale across complex domains. As a long-term healthcare practitioner with experience operating within HIPAA regulatory frameworks as a licensed clinical provider, I built within compliance frameworks before architecting beyond them — and FSL's infrastructure operates outside HIPAA scope by architectural design, holding no PHI. As founder of Future Systems Lab, I move from concept to production: on-chain deployments, EIP-191 wallet-gated consent architecture, and infrastructure designed to operate outside HIPAA scope by holding no PHI — a stronger architectural position than compliance, because the regulation doesn't apply when the system never holds the protected data. I think in systems, across disciplines, at the intersection of governance and user experience. Nine Ethereum smart contracts deployed and verified on Sepolia — every FSL system is in production. Pursuing a Doctor of Engineering to formalize applied research methodology and advance systems leadership at the intersection of blockchain, AI, and decentralized governance.

## ENGINEERING EXPERIENCE

### Lead Engineer — SovereignLedger Sovereign Health Governance Infrastructure

2026 – Present

Sepolia: [0x4afA577fA914068451e0Aa97b61F23960f02aCc4](#)

- Engineered a full PostgreSQL session governance schema spanning participants, sovereign guides, sessions, attestations, documents, audit logs, consent grants, and compliance log tables.
- Built Phase 3 SMS-based approval-gated automation with case-insensitive approve/deny/defer, 4-hour timeout, and max-defer logic — deployed to production VPS via systemd.
- Architected SovereignLedger as a standalone blockchain governance layer for session governance, attestation, and participant-controlled record verification — integrates with any payment pipeline (Crypto → XRP → Sovereign Navigation) and deployable independently of any single platform.

### Practice Operations Lead — Riccobene Associates

2019 – 2025

70+ office practice network, North Carolina

- Led operational scaling across 70+ office practice network, producing 64% per-provider production growth (\$380,710 → \$624,640 average per provider) through team coordination, workflow optimization, and cross-location process standardization.
- Designed and deployed two-tier production-share compensation model adopted company-wide — 27% standard tier (network-wide rollout) and 33% architect tier (system designer compensation).
- Per-provider share of total office production grew from 30.4% to 38.1% — system redesign enhanced overall office economics rather than redistributing existing revenue.
- Recruited directly by CEO for C-suite engagement; maintained multi-year engagement while concurrently founding Future Systems Lab.

### Lead Engineer — SovereignSession On-Chain Session Attestation

2026 – Present

Sepolia: [0xbeb13A360C6F0C77Ea3af3650Ab9762a1B9965A1](#)

- Designed and deployed wallet-native session attestation contract operating outside HIPAA scope by holding no Protected Health Information. EIP-191 wallet authentication for Sovereign Guide and Participant — no name fields, no PII captured by FSL infrastructure.
- On-chain SessionStarted and SessionCompleted events with timestamps and duration only — immutable proof that a session occurred between two consenting wallets without revealing session content.
- Phase 5 doctoral research (post-ASU acceptance): Two-Party Wallet-Signed Mutual Authentication with Client-Side Encrypted Session Recording. Advances SovereignSession from single-party attestation (guide-initiated, deployed Phase 1–4) to bilateral cryptographic consent — both Sovereign

Medical Neuroscience — Duke SOMApr 2025

Health Informatics on FHIR — Georgia TechMar 2025

Business Analyst — IBMJun 2025

Data Analytics · UX Design · PM — GoogleJun 2025

IPFS · Filecoin · libp2p · Merkle DAGsIPFS / Filecoin Foundation / ProtoSchool · 2025

NC Accident & Health Insurance ProducerPre-licensing completed Apr 2026, exam scheduled

National Provider Identifier (NPI):  
1497696264Naturopath (175F00000X), NPPEs, 2026

Guide and Participant submit EIP-191 signatures before session start, verified on-chain. Client-side AES-256-GCM encrypted video, IPFS-pinned storage, on-chain hash proof, participant-controlled decryption keys. No competitor (Doxy.me, Daily.co, Zoom) implements two-party wallet-signed mutual consent with client-side encrypted recording and on-chain attestation.

## Lead Engineer — AlchemistForge Permissionless On-Chain Shadow Integration Primitive

2026 — Present

Sepolia: [0xE092336F8f5082e57CcBb341A110C20ad186A324](#) · [alchemyforge.io](#)

- › Designed and deployed a fully permissionless smart contract for voluntary on-chain Jungian shadow integration — no admin, no owner, no pause mechanism, no token economics. Architecture covered by U.S. Provisional Patent 64/063,037.
- › Two-function design: `alchemyize()` records voluntary shadow disclosure; `celebrate()` marks integration completion. Both emit publicly queryable events. Zero PII stored on-chain — only pseudonymous wallet address, voluntary self-authored text, and timestamps.
- › Live deployment integrates Brave Wallet + EIP-191 signature for participant authentication, public Blockscout verification, and "verify your contribution" UX inviting any visitor to inspect their own on-chain record.
- › Novel celebration animation: Ophiuchus (suppressed 13th zodiac) traverses celestial firmament with trailing sparkle stream while the 12 standard zodiac glyphs burst from center — symbolic representation of integrated shadow material surfacing.
- › Scaffolded mental-health-on-chain as a five-entry public smart contract awareness series; deployed 24/7 campaign orchestration across Dev.to, Twitter, Discord, GitHub, and Telegram. Recorded on-chain case study as Participant #1.

## Founder & Systems Architect — Future Systems Lab

2023 — Present

[github.com/Future-Systems-Lab/Future-Systems-Lab-profile](#)

- › Deployed and verified nine Ethereum smart contracts on Sepolia testnet: [HypnoNeuroToken \(HNT\)](#), [EHTv2](#), [MindMasteryNFT](#), [SovereignLedger](#), [AlchemistForge](#), [BenevolenceFund](#), [SovereignAchievement](#), [SovereignSession](#), and [NeuroBalanceConsent](#).
- › Architected a 17-agent Master Command Center spanning Audit, Smart Contract, Backend Build, Frontend/UI, Compliance, Security, Testing/QA, and Research agents with full dropdown panel orchestration.
- › Deployed production VPS automation stack (clawdbot) on OpenClaw VPS: Flask approval server, Telegram bot, Streamlit analytics dashboard, and two systemd services running continuously.
- › Architected EIP-191 wallet-signed consent as unified authentication primitive across all 5 FSL platforms — single cryptographic signature serves as identity verification, informed consent, and session authorization. This integrated pattern is covered by U.S. Provisional Patent 64/063,037 and is the subject of a manuscript in preparation for Blockchain in Healthcare Today.
- › Engineered EncryptAI — a 19-step canonical monorepo (frontend/, backend/, labs/, contracts/, game/) with Brave Wallet authentication via window.ethereum + EIP-191, PostgreSQL 17, and blockchain writes via viem — [Step 2 deployed as Secure Health Login](#).
- › Established SSH-authenticated signed commits, GitHub org governance, and Rights Reserved licensing policy across all FSL repositories.

## Healthcare Data Architect — HypnoNeuro Mental Wellness Platform 2023 — Present

*HypnoNeuro · EncryptHealth · SovereignLedger · NeuroBalance Watch — modular systems engineered to operate independently or as an integrated ecosystem*

- › Architected and deployed a tier-based therapeutic game framework — currently operating with three flagship demos (one per level: L1 Theta Wave/GABA, L2 Biochemical/Serotonin, L3 Original Frequency/Endocannabinoid) demonstrating the full participant flow: wallet gating, HNT token rewards, on-chain achievement minting, and NFT-based tier access. Architecture supports expansion to full catalog and roadmap targets VR-mediated avatar circle integration.
- › Architected dual-chain healthcare session anchoring — every session cryptographically committed to both Ethereum Sepolia and XRPL testnet, generating a tamper-evident audit trail for sovereign health

- records.
- › Built ISOM-aligned 4-tier provider credentialing system (Platinum/Gold/Silver/Verified) with automated license verification via Verifiable.com and AI-assisted credential review using Claude Vision.
  - › Implemented Walsh Protocol AI recommendation engine — biotype classification across 5 neurotransmitter systems (Trp, Tyr, Cortisol, Dopamine, Epi/NorEpi) with plant-based nutrient protocol generation.
  - › Deployed avatar-based anonymous session rooms via Daily.co WebRTC — wallet-connected, provider-facilitated video sessions with sessionStorage-persisted character customization and SovereignLedger billing anchor.
  - › Designed white-label operator licensing model with three commercial tiers (Starter \$497/mo, Growth \$1,497/mo, Enterprise custom) — full registration flow, Telegram-notified applications pipeline, and operator config abstraction layer.
  - › Completed 35-step canonical build across 7 sprints — all five platforms live on Vercel and interconnected: HypnoNeuro, EncryptHealth, SovereignLedger, NeuroBalance, Provider Portal, and FSL Command Center.
  - › **Sprint 1–3:** Core platform buildout — therapeutic game framework, Brave Wallet + XRPL dual-wallet auth, PostgreSQL schema, FHIR R4 export, XRPL session anchoring, NFT gating, HNT token rewards, provider portal backend.
  - › **Sprint 4–5:** Avatar Session Rooms (Daily.co WebRTC), AlchemistForge HNT bridge, endocannabinoid baseline ML scorer, COA Lookup, Support Circles, Walsh Protocol recommendation engine, wellness scoring, XRPL wallet connect.
  - › **Sprint 6:** XRPL wallet polish (GemWallet/Crossmark/Xaman/manual), demo mode with watermark and seeded leaderboard, biosensor infrastructure spec (5 target hardware partners), FSL Command Center on Vercel, Telegram operator notification pipeline.
  - › **Sprint 7 (current):** Intake form Walsh Protocol biotype pre-screening, win screen confetti + NFT progress tracking + share flow, support circle categories and HNT bonus, ecosystem health monitoring (cron + Telegram alerts), OSS portfolio formalization.
  - › Planned hardware integration: NeuroBalance biosensor layer targeting Nutromics (sweat biomarkers), Polar (HRV), OpenBCI (EEG), Emotiv, and Dexcom (CGM) — full API spec and device schema documented.
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## Naturopathic Psychology Practice

2018 – Present

*Dr. Margarita Davenport D.N.Psy · Naturopathic Psychology and Hypnosis Center · NPI 1497696264*

- › Founded and operated independent clinical practice applying naturopathic psychology and orthomolecular frameworks — amino acid therapy protocols (Linus Pauling, Abram Hoffer, William Walsh, Albert Mensah), integrative behavioral health, and sovereignty-first participant engagement.
  - › Operated SunCodes Holistic Health (2018–2024): developed seed-to-shelf neurotransmitter-targeted CBD wellness line, managed full product lifecycle (formulation, COA testing, GMP documentation), and designed therapeutic wellness center buildout.
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## Earlier Clinical Operations Leadership

1996 – 2019

- › Progressive leadership across multi-location dental and healthcare practice operations. Roles spanning clinical coordination, practice management, multi-site team coordination, and operational scaling — foundation for the workflow redesign and compensation architecture work later led at Riccobene Associates.
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## OPEN SOURCE CONTRIBUTIONS

### Open Source Contributor — Healthcare & Blockchain

Jul 2025 – Present

*Self-Directed / Independent*

- › [DIF / Presentation-Exchange #486](#) — OIDC4VP healthcare consent flow example · Open, awaiting named review by Daniel Buchner (csuwildcat), Kim Hamilton Duffy (kimdhamilton), Brent Zundel, and rado0x54 · Foundation for SovereignLedger's consent architecture and EIP-191 wallet authentication
- › [OpenZeppelin Contracts #6414](#) — ReentrancyGuard documentation clarification · Open · FSL contracts inherit from OpenZeppelin audited base contracts
- › [Hyperledger Fabric #5419](#) — Build and runtime environment notes · Open · Permissioned ledger patterns for FSL provider credentialing

- › [Ethereum / EIPs #11409](#) — Markdown formatting fix to README's EIP document structure section (Abstract / Motivation / Specification / Rationale / Security Considerations) · Open, in active review · Referenced in EIP Editor Office Hours #93 (Mar 17, 2026) and #94 (Mar 24, 2026); all 9 automated checks passed · Improves rendering consistency for the constitutional document of the EIP standards process
- › [Uniswap v3-core #1077](#) — Invariant test note for pool math · Open · Reference for future HNT/ETH liquidity pool design
- › [Balancer v2 #2648](#) — Pool math reference documentation · Open · HNT liquidity pool and AlchemistForge bridge architecture reference
- › [BigchainDB #2763](#) — Python quickstart documentation · Open · Immutable data patterns informing NeuroBalance biosensor anchoring
- › [OpenMRS / FHIR2 #594](#) — Quickstart pointer for healthcare interoperability · Closed · Same FHIR2 module stack as EncryptHealth FHIR R4 export

*Applied projects:*

- › [Secure Health Login](#) — practitioner-facing smart wallet authentication app implementing EIP-712, DID/OIDC, and sovereignty-first access control — live on Vercel

## PUBLICATIONS

### BLOCKCHAIN INFRASTRUCTURE

#### "Ending the Insurance Verification Nightmare"

America Out Loud · March 2026 · [americaoutloud.news](#)

Proposed blockchain governance as a systemic solution to claims fraud through decentralized eligibility verification and immutable audit trails.

### DIGITAL HEALTH & BEHAVIORAL WELLNESS

#### "Reprogram Your Limiting Beliefs on the Blockchain While Learning Web3"

America Out Loud · April 2026 · [americaoutloud.news](#)

Introduced AlchemistForge — Jungian shadow integration permanently recorded on Ethereum. Demonstrates behavioral health as a proving ground for decentralized data governance.

### GOVERNANCE & CONSENT ARCHITECTURE

#### "Blockchain Consent Transforms Patient Data Control"

America Out Loud · March 2026 · [americaoutloud.news](#)

Examined how smart contract-governed consent flows resolve systemic failures of centralized health data management — the architectural basis for EIP-191 wallet-signed consent.

*Full archive (weekly contributor, 2024–present): [americaoutloud.news/author/meg-montanez-davenport](#)*

## RESEARCH

### Intellectual Property

U.S. Provisional Patent Application No. 64/063,037 · Filed 11 May 2026 · Sole Inventor

"System and Method for Sovereign Data Governance via Wallet-Signed Consent Attestation in a Zero-PHI Behavioral Health Architecture." Future Systems Lab LLC.

### EIP-191 Consent Architecture for Sovereign Behavioral Health Data Governance — Lead Author

Manuscript in preparation; target journal: Blockchain in Healthcare Today (BHTY)

Establishes the EIP-191 wallet-signature consent pattern as a generalizable architecture for sovereign data governance, with regulated behavioral health as the deliberate stress-test domain. Documents the implementation deployed in SovereignLedger v2 (Sepolia:

[0x4afA577fA914068451e0Aa97b61F23960f02aCc4](#)).

## EDUCATION

BS, Management Information Systems  
CSU Global — Regionally Accredited (HLC)

Conferred July 2026

Doctor of Naturopathic Psychology  
Kingdom College of Natural Health

Conferred

## ENTITY & INTELLECTUAL PROPERTY

**Future Systems Lab LLC** (Wyoming) — EIN 42-2050630

**Trademark:** USPTO Serial No. 99533250 — "Future Systems Lab," Class 42 (Software/Tech Platform) — Publishing May 19, 2026

**Trademark:** USPTO Serial No. 99821948 — "Future Systems Lab," Class 35 (Online Marketplace/Directory) — Filed May 13, 2026

**Patent:** U.S. Provisional Application No. 64/063,037 — "System and Method for Sovereign Data Governance via Wallet-Signed Consent Attestation in a Zero-PHI Behavioral Health Architecture" — Filed May 11, 2026 — Assigned to Future Systems Lab LLC (Assignment ID 1803665)

National Provider Identifier (NPI): 1497696264 — Naturopath (175F00000X), NPPES, 2026

PROPOSED APPLIED RESEARCH

### **Decentralized Sovereign Data Governance: Behavioral Health as a Proving Ground for Wallet-Signed Consent Architectures**

Phases 1–4 deployed and tested: EIP-191 wallet-signed consent, zero-PHI architecture, nine verified Sepolia smart contracts, five-platform ecosystem. Phase 5 capstone research: **Two-Party Wallet-Signed Mutual Authentication with Client-Side Encrypted Session Recording** — advances SovereignSession from single-party attestation to bilateral cryptographic consent (both Guide and Participant submit EIP-191 signatures, verified on-chain), with client-side AES-256-GCM encrypted video, IPFS-pinned storage, on-chain hash proof, and participant-controlled decryption keys. No existing telehealth platform (Doxy.me, Daily.co, Zoom) implements two-party wallet-signed mutual consent with client-side encrypted recording and on-chain attestation.

| [SovereignSession on Blockscout](#)